COMPUTER VISION (ITA0505)

Assignment 1 – Slot A

Design an Image Acquisition and Analysis Front-End for High-Speed Textile Fabric Inspection

Case: WeaveSure Textiles – cotton-polyester fabric web, 1,600 mm wide, 0.75 m/s

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Summary of decisions

Task	Question	Recommended choice (from the evidence in this report)
1	Inspection approach	C – automated vision for continuous running, with image-assisted review of flagged faults; manual alone cannot hold the 2% miss target
2	Pipeline depth	C – three-level (regions → features → Random-Forest decision): 26/30 faults, 0/30 false alarms vs 25/30 and 8/30 for rule-only
3	Colour representation	CIE Lab (a*, b*, ΔE) with a white-reference correction; HSV only as a quick brightness-tolerant check
4	Storage	Original = lossless PNG (never overwritten); JPEG only for display copies
5	Acquisition	Line-scan camera, encoder-timed, exposure ≤0.27 ms, DC LED lighting
6	Resolution	8,192 px across (0.195 mm/px); upgrade path to 12,288 px
7	Bit depth	Capture 10/12-bit, flat-field in that depth, store 8-bit evidence; do not go below 8-bit
8	Optics + lighting	B – enclosed diffuse LED, calibrated f≈24 mm lens, flat-field correction
9	Front-end	Configuration B (balanced); first upgrade if miss target fails = resolution/second camera
10	New requirement	MODIFY (staged): new lens for 1,800 mm, shorter exposure, back-light for micro-holes; REDESIGN to a spectral channel only if the contamination test fails

Task 1 – Overview and Features of Computer Vision [4 marks]

1.1 Study design

Dataset (simulated): 60 evaluation images of 256×256 px (about 33×33 mm each at 0.13 mm/px): 30 normal (8 plain, 8 with a thick yarn, 7 with a fold shadow, 7 twill) and 30 faulty (6 each of hole, broken yarn, slub, oil mark, shade band). A separate set of 80 images was used only to set thresholds and train the classifier, so the test images were not used for tuning.

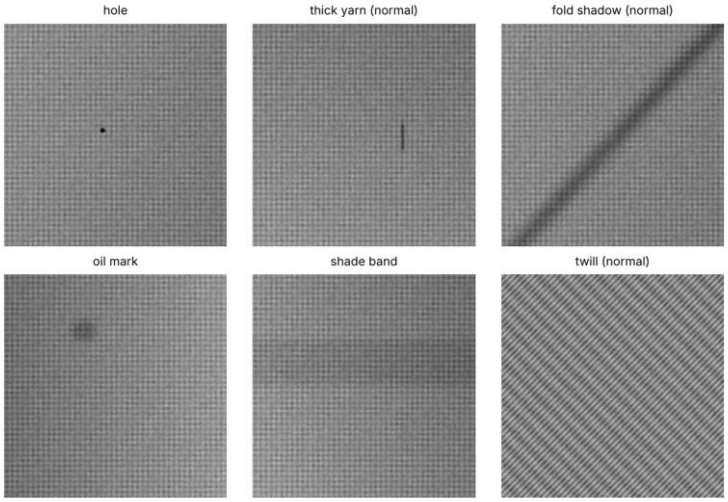


Figure 1 – Example images: faults (hole, oil, shade band) and normal look-alikes (thick yarn, fold shadow, twill).

Approach A (manual): three evaluators, two rounds each, on all 60 images (simulated raters; each rater's detection probability depends on fault type, with a shared "image difficulty" factor and a personal factor, so agreement is realistic but not perfect). **Approach B (image-assisted):** the operator sees CLAHE-enhanced, magnified regions; modelled with the same rater simulation but with higher detectability (assumed). **Approach C (automated):** the rule-based and classifier prototypes of Task 2 (OpenCV code in Appendix A).

> **Prediction (before testing):** Manual inspection will catch obvious holes but miss low-contrast shade and oil faults; image assistance will raise recall at the cost of more false alarms; the rule-based vision prototype will be the most repeatable but will confuse thick yarns with faults.

1.2 Results

Approach	Faults found (of 30)	False alarms (of 30 normal)	Time / decision
A. Manual (3 raters × 2 rounds, simulated)	84.4% (mean)	8.9%	Operator time; ~1 s of attention per window (assumed)
B. Image-assisted (simulated operator)	93.3%	16.1%	Operator time + enhancement ≈ 1 ms/image (measured)
C1. Automated, rule-based (Task 2 pipeline B)	25/30 = 83.3%	8/30 = 26.7%	5.3 ms/image (measured, 1 core)
C2. Automated + classifier (Task 2 pipeline C)	26/30 = 86.7%	0/30 = 0.0%	12.6 ms/image (measured, 1 core)

Inter-rater and intra-rater agreement (Cohen's kappa, simulated raters, fault / no-fault decision on 60 images): inter-rater κ (mean of 6 rater-pair/round combinations) = **0.67** (range 0.63–0.73); intra-rater κ (round 1 vs round 2, per rater) = 0.73, 0.83, 0.83 (mean **0.80**). On the common Landis–Koch scale 0.61–0.80 is "substantial": raters are fairly consistent with themselves but disagree with each other on ~1

image in 6, mostly the low-contrast faults. For comparison, the assisted raters reach inter-rater κ = 0.82. An automated rule gives κ = 1.0 with itself by construction (deterministic).

1.3 Comparison table

Criterion	A. Manual	B. Image-assisted	C. Automated vision
Inspection rate	≈10–20 m/min (assumed typical)	≈20–30 m/min (assumed)	Line speed 45 m/min; needs ≈31 Mpx/s vs ≈25 Mpx/s per core measured → needs 4 cores
Repeatability	κ≈0.67 between people; drifts with fatigue	Better (κ≈0.82) but still human	Deterministic; same input → same output
Objectivity	Subjective, trained eye	Partly objective	Explicit numeric thresholds
Non-contact	Yes	Yes	Yes
Fault-location accuracy	±several cm (eye, marker)	±1–2 cm (screen)	≈0.05–0.1 mm in the image; limited by encoder/marker (see Task 9)
Continuous operation	No (breaks, shifts)	Partly	Yes (24 h)
Traceability	Paper note	Screenshot possible	Image + coordinates + rule version stored
New fault type	Training of staff, days	Days	Re-tune rules or relabel & retrain (hours–days)
Operating cost	Wages recur	Wages + screen	Capital + maintenance; low recurring

1.4 Fatigue over a 10-hour shift

Human vigilance falls with time on task (the "vigilance decrement" first reported by Mackworth in the 1940s). I used an **assumed** decay model, detection = start value × (0.70 + 0.30·e^(−t/2 h)), to show the likely scale. With the simulated start detection of 84%, the 10-hour average falls to about **64%** and the last hour to 59%, i.e. roughly a third of faults are missed late in the shift, against the 2% target. The parameters are assumptions; a real study would log detection against shift hour.

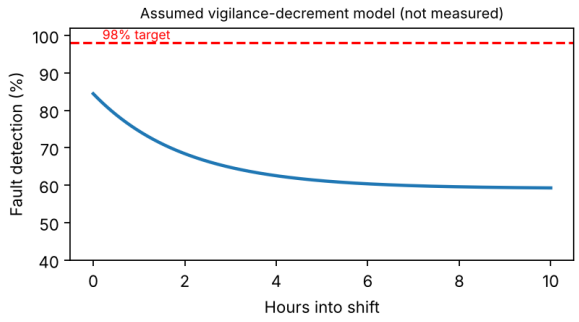


Figure 2 – Assumed detection decay over a 10-hour manual shift.

1.5 Image processing, computer vision, machine vision, computer graphics

Field	Input → Output	Place in the fabric line
Image processing	Image → image (flat-field, smoothing, threshold mask)	Low-level clean-up right after the sensor
Computer vision	Image(s) → description/decision ("oil mark, 6 mm")	Fault interpretation and grading
Machine vision	Image + hardware (lights, camera, encoder, I/O) → action (mark, log, stop)	The whole installed inspection station
Computer graphics	Description/model → image (render, annotation overlay)	Operator display, annotated copies, synthetic test images (as used here)

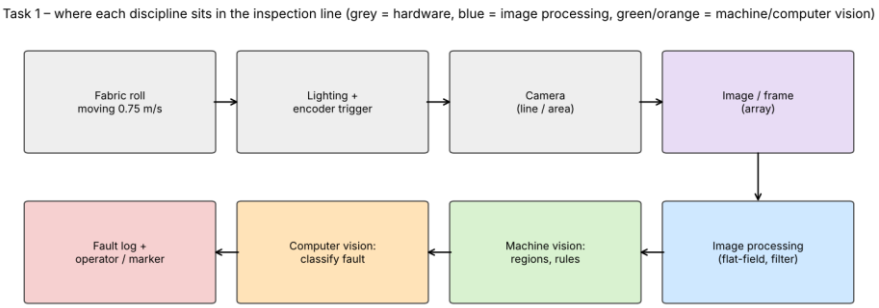


Figure 3 – Workflow diagram of the inspection line.

1.6 Unseen faults and the review procedure

A rule-based system only knows what its rules describe. A new fault (for example a coating streak) either falls inside an existing rule and is **mislabelled**, or matches nothing and is **silently accepted as normal**. Task 2 showed both risks: the thick-yarn rule fired on every normal thick yarn (8/8), while the classifier missed 3 of 6 broken yarns whose measurements overlapped with thick yarns. Recommended procedure: (1) make the decision three-way – *normal / known fault / unknown anomaly* – by sending any region that is statistically unusual but matches no class to a review queue; (2) the operator labels queue items daily; (3) once ≥30 examples of a new type exist, add a class, retrain, and test on the stored evidence set before release; (4) every rule/model change gets a version number stored with each inspection record; (5) track the accept rate of the review queue weekly – a sudden change flags drift.

1.7 Recommendation and limits

Automated vision (C) is the most suitable approach for continuous inspection, with image-assisted review (B) for flagged faults and for labelling unknowns. Limits: the prototype was tested on synthetic stills, not on a moving web; its best simulated result (86.7% recall, 0% false alarms on this set) does **not** reach the 2% missed-defect target, and the rule-only version raises many false alarms. It must not be called production-ready until tested on a real web with real fabric.

Task 2 – Levels of Computer Vision and Image Processing [5 marks]

Same 60 test images and the same separate 80-image calibration set for all three pipelines. Code: Appendix A. Stage inputs/outputs: image → (grayscale, smoothing, threshold) → mask → (morphology, connected components) → connected regions → (area, length, width, orientation, contrast) → feature vector → (rule or Random Forest) → final label.

Prediction (before testing): C (classifier on measurements) ≥ B (fixed rules) > A (dark-pixel count). A will raise false alarms on thick yarns; B will confuse thick yarns and broken yarns.

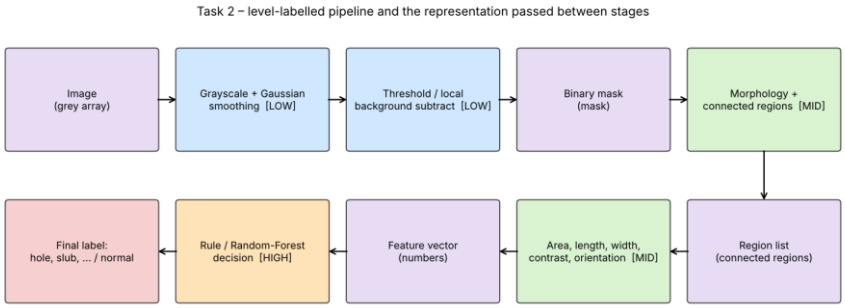


Figure 4 – Level-labelled pipeline and the representation passed between stages.

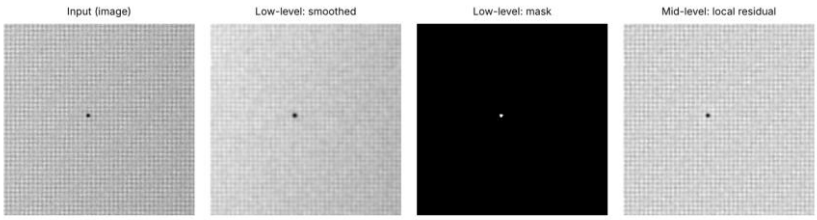


Figure 5 – Stage outputs on a hole image: input, smoothed, mask, local residual.

Measure (60 test images)	A. Low-level	B. Low+mid (rules)	C. Three-level (RF)
Faults detected (of 30)	4	25	26
False alarms (of 30 normal)	0	8	0
Accuracy	57%	78%	93%
Thick yarn flagged (of 8)	0	8	0
Fold shadow flagged (of 7)	0	0	0
Twill flagged (of 7)	0	0	0
Holes / broken yarn / slub / oil / shade found (of 6 each)	4 / 0 / 0 / 0 / 0	6 / 1 / 6 / 6 / 6	6 / 3 / 6 / 6 / 5
Time per 256×256 image (1 core, measured)	0.13 ms	5.3 ms	12.6 ms
Explainability	Very high	High (each rule readable)	Medium (feature importances)
Effort to add a new fault	Re-tune two numbers; cannot separate types	New rule by hand	Label ≥30 examples, retrain

Outcome vs prediction. The order A < B < C held for correct faults and for false alarms. A surprise: pipeline A produced **zero** false alarms but found only 4/30 faults – its threshold was tuned on the calibration set to avoid thick-yarn alarms, which made it blind to everything except holes. The hard cases were as expected: thick yarns (8/8 false alarms in B), and broken yarns, which C found in only 3/6 because their length/width/contrast overlap with thick yarns.

Why a dark-pixel rule calls a thick yarn a hole, and how to improve it

A thick yarn is a locally dark, elongated blob; a dark-pixel counter only sees “many dark pixels in the window”, so a 3×40 px thick yarn and a hole look alike. A region measurement separates them: a hole is compact (aspect ratio ≈1, high local contrast, surrounded by regular texture), a thick yarn is long and straight (aspect >8) with moderate contrast, and the surrounding texture continues along its length. Poor measurements can still mislead the final decision: if blur, a low threshold or an overlapping fold merges two regions, the length/width/contrast the classifier receives are wrong, and even a perfect classifier then gives the wrong label. This is the cause of the 3/6 broken-yarn result above (garbage-in, garbage-out across levels).

Staged pipeline vs end-to-end CNN / Vision Transformer

A CNN or ViT learns features directly from pixels and generalises better to varied textures, but typically needs thousands of labelled images **per fault class**, rare defects included, and GPU hours for training; a ViT needs more data and compute than a small CNN. With no GPU and a 4-core PC, training on-site is impractical; inference of a small CNN on 8,192-px-wide lines at 31 Mpx/s would also strain the CPU. The staged pipeline needs only tens of examples, runs in ~14 ms per 256×256 tile here, and every decision can be explained to a quality inspector. Recommended: **pipeline depth C (staged, CPU-only)**; consider a small CNN later only on the cropped regions it flags, trained off-site.

Task 3 – Digital Image Fundamentals I: Colour Representations [4 marks]

Data (simulated): 60 fabric patches (64×64 px, BGR, uint8) – 3 shade groups (Lab targets about ΔE≈5 apart) × 3 lighting conditions × 10 patches, with fibre texture, dye-batch variation and glossy streaks. L1 = reference, L2 = same colour of light but 35% dimmer, L3 = dimmer **and** colour cast (night LED, 0.62/0.66/0.72 gains in B/G/R). Light scaling was applied in linear light. Same sample IDs (e.g. 2L3-07) are used for all methods. Classifier: nearest centroid trained on L1 only.

Prediction (before testing): Lab will separate the shade groups best and be least affected by brightness; HSV will be good for brightness changes but fragile near hue wrap-around; RGB will fail when brightness changes.

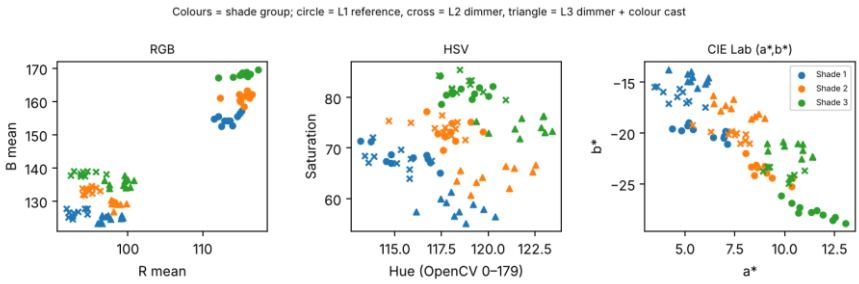


Figure 6 – Shade groups in RGB, HSV and Lab (a,b*) space for three lighting conditions.*

Representation	Same light (leave-one-out)	Test on L2 (dimmer)	Test on L3 (dimmer + cast)	L2/L3 after white-reference correction	Time/patch
RGB/BGR (mean B,G,R)	100%	33%	33%	98%	0.08 ms
HSV (hue, saturation; weak+bright pixels rejected)	93%	100%	33%	92%	0.09 ms
CIE Lab (L*,a*,b*; ΔE76)	100%	33%	33%	98%	0.25 ms
CIE Lab (a*,b* only; chroma ΔE)	100%	33%	33%	98%	0.22 ms

Outcome vs prediction (differs in part). All representations separate the groups well under one light (93–100%). Under a pure brightness change only **HSV** survived (100%), as hue and saturation are brightness-invariant; RGB and Lab (L*, a*, b*) collapsed to chance (33%) because L* and the colour channels move with brightness. Under a colour cast every representation failed uncorrected. After dividing by a **white-reference tile** (measured with 1% noise), RGB and Lab recover to 98% and HSV to 92%. So my prediction that Lab alone would be most robust was wrong: the **white reference matters more than the colour space**; Lab is preferred afterwards because ΔE has a physical meaning (about 1 ΔE is a just-noticeable difference) and is easy to tell an inspector ("this roll is ΔE 3.2 from the master"). Overlap: with ΔE≈5 between groups and gloss, the uncorrected distributions of L2/L3 sit entirely inside other groups' regions (Figure 6).

Comparison points

Aspect	RGB/BGR	HSV	CIE Lab
Group separation (same light)	Good	Good (needs S,V masking)	Good; ΔE shows how far
Brightness sensitivity	High	Low for H,S	High for L*; a*,b* change slightly
Glossy fibres	Raise all channels	Mask V>230 and S<40	Median of L*,a*,b* reduces effect

Aspect	RGB/BGR	HSV	CIE Lab
Hue wrap-around	None	Yes – red hue near 0 and 179 in OpenCV; use circular mean (done in code)	None
Speed	0.05 ms	0.09 ms	0.2 ms (conversion)
Explaining to an inspector	Hard	Medium	Easy (ΔE)

Image-array properties and indexing

For patch 1L1-00: **shape = (64, 64, 3)**, **dtype = uint8**, **channels = 3**, nbytes = 12288. Pixel at row 10, column 20 = **[157,128,116] in B,G,R order**, which is RGB [116,128,157].

NumPy indexes as **img[row, column]** = img[y, x]: the first index is vertical. Geometric coordinates, OpenCV drawing functions (cv2.circle, cv2.rectangle) and cv2.resize's size argument use **(x, y)** = (column, row). Mixing them transposes a fault's position – a 12 mm error in the wrong direction on this web.

Storage: b = M × N × k bits. For one 8,192 × 4,096 frame, 8-bit mono: 8,192 × 4,096 × 8 = 268,435,456 bits = 33.55 MB. As 3-channel 8-bit (k = 24): 100.7 MB. Patch above: 64×64×24 = 98,304 bits = 12,288 bytes.

BGR read as RGB: red and blue swap, so a blue-dyed fabric appears orange; hue values shift by ~120°, Lab a*/b* change sign – the shade grouping and any reference colour comparison silently become wrong although nothing crashes.

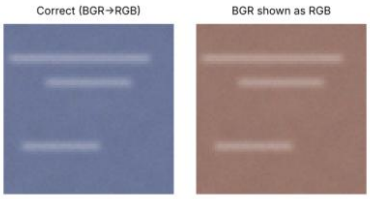


Figure 7 – Correct BGR→RGB (left) vs BGR displayed as RGB (right).

Colour representation vs a pretrained-model embedding. A pretrained network turns the patch into a 512–2048-number vector that can separate shades well in a learned space, but individual numbers have no physical meaning and the result changes with the model version. Lab/ΔE gives a **more transparent explanation for shade grading**: a single number tied to a reference sample and an agreed tolerance. **Selected: CIE Lab with white-reference correction.**

Task 4 – Digital Image Fundamentals II: Storage Representations [4 marks]

Data (simulated): 20 colour fabric images, 1,024 × 1,024 px, 3 channels, 8-bit (uint8), each containing several holes, gaps, slubs, oil and shade faults. Each was saved as raw NumPy (.npy), PNG (compression 6) and JPEG at quality 95, 75 and 50; times are on a 1-core test machine.

Prediction (before testing): PNG will return identical pixels; JPEG will keep large faults but soften one-pixel gaps and small holes, and repeated saving will accumulate damage.

Format	Dimensions / channels / depth	Mean file size	Write	Read	PSNR vs original
A. Raw NumPy array (.npy)	1024×1024, 3 ch, 8-bit uint8	3.15 MB	3.7 ms	0.7 ms	∞ (identical)
B. Lossless PNG	1024×1024, 3 ch, 8-bit uint8	1.45 MB	242.0 ms	25.6 ms	∞ (identical)
C1. JPEG quality 95	1024×1024, 3 ch, 8-bit uint8	0.40 MB	4.9 ms	6.2 ms	40.4 dB
C2. JPEG quality 75	1024×1024, 3 ch, 8-bit uint8	0.14 MB	3.7 ms	3.8 ms	37.0 dB
C3. JPEG quality 50	1024×1024, 3 ch, 8-bit uint8	0.10 MB	3.0 ms	2.9 ms	35.8 dB

Decoded-array properties were identical for all formats (shape (1024,1024,3), uint8); only JPEG changed values. PNG decoded **bit-identical: true**. Contrast of a 1-pixel dark gap (70 grey levels original): PNG 70.0, JPEG q95 70.1, q75 69.8, q50 68.6. A 4×4 px hole (80 grey contrast): q95 79.5, q75 79.0, q50 **69.3**. A 6-grey-level step stayed 6.0 at all qualities.

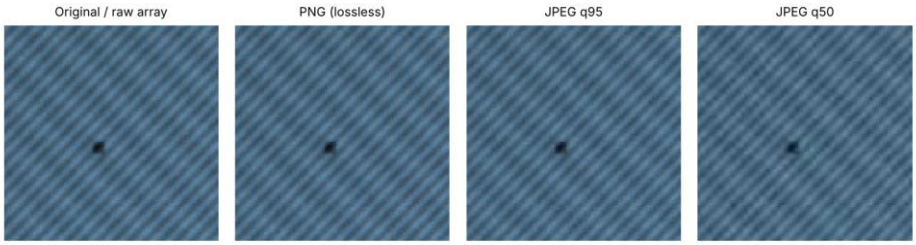


Figure 8 – 100% crops (96×96 px) of a fault area: original, PNG, JPEG q95, JPEG q50.

Outcome vs prediction. Largely as predicted: PNG was exact and about 2.2× smaller than raw on these synthetic images (real fabric compresses less because of sensor noise; test with real frames). JPEG q95 is visually safe for the isolated defects tested here but the hole contrast already drops at q50 (80 → 69), and JPEG changes pixel values, so measurements are no longer reproducible from the file. One difference: the 1-pixel gap lost very little contrast in a *flat* background; in real yarn texture JPEG block artefacts and 4:2:0 chroma subsampling hurt more – I did not test that, so treat the q75/q95 numbers as optimistic.

Dimensions vs channels vs bit depth vs compression

Dimensions = M × N pixels (here 1,024×1,024). **Channels** = values per pixel (1 grey, 3 colour). **Bit depth** = bits per value, giving 2^k intensity levels (8 bit = 256). **File compression** is how the numbers are encoded in the file. **Lossless compression** (PNG) stores the same numbers in fewer bytes and decodes to exactly the same array. **Reducing the number of intensity values** (8 → 4 bit, Task 7) throws information away *before* storage and cannot be undone by any decoder; JPEG is lossy compression that changes values through its transform quantization.

Storage calculation and 7-day estimate

Assumptions: recommended 8,192-px line-scan, 0.195 mm/px, lines every 0.195 mm at 0.75 m/s = 3,840 lines/s, 8-bit mono, 24 h × 7 days = 604,800 s; one fault per 20 m of fabric (assumed, to be replaced by plant data).

Quantity	Calculation	Result
One frame 8,192 × 4,096 × 8-bit	8,192 × 4,096 bytes	33.6 MB uncompressed
Continuous raw data rate	3,840 lines/s × 8,192 B	31.5 MB/s
7 days, every line kept raw	31.5 MB/s × 604,800 s	19.0 TB (9.5× the 2 TB limit)
Evidence policy – fault windows	0.75 m/s × 86,400 s ÷ 20 m ≈ 3,240 faults/day; 8,192 × 2,048 window = 16.8 MB raw → ≈ 8.4 MB PNG (≈2:1)	≈ 27 GB/day → 190 GB / 7 d
Periodic full-resolution reference frame	1 per minute × 17 MB PNG	≈ 24 GB/day → 170 GB / 7 d
Continuous low-res record (÷8, JPEG q85 ≈ 10:1)	≈ 0.48 MB/s raw → ≈ 0.05 MB/s	≈ 30 GB / 7 d
Policy total		≈ 0.39 TB, about 20% of 2 TB

JPEG generation loss; annotation must not overwrite evidence

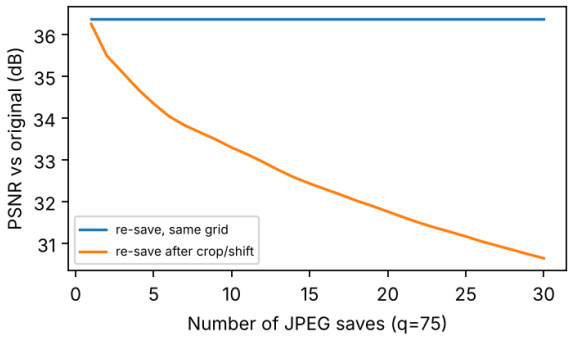


Figure 9 – Repeated JPEG saving (quality 75) of the evidence image.

Re-saving at the same quality and alignment is almost idempotent (PSNR stays 36.4 dB), but real workflows re-save after cropping, resizing or annotating, which shifts the 8×8 block grid. With a 3-px shift between saves PSNR falls from 36.2 dB after save 1 to 33.3 dB after 10 and 30.6 dB after 30 – fine yarn gaps blur further each time. The annotated copy (a red box drawn on the image) differs from the original (identical = false); if saved over the original, the box destroys the very pixels an auditor needs and JPEG compounding follows. Annotations belong on a separate display copy generated on demand.

Recommended storage policy

- **Original evidence record:** the lossless PNG of the captured fault window (plus its encoder position, timestamp, camera settings and rule version in a JSON sidecar), written once and set read-only; hash stored in the log.
- **Working array:** raw NumPy in memory only (fast, no I/O); **display copy:** JPEG q85–90 annotated, regenerable.
- **Retention:** 7 days, ≈0.4 TB; retrieval of one fault window = one PNG read (~25 ms/MB-class on test machine, fine for review).
- If auditors insist on every line at full resolution, 19 TB is needed – outside the 2 TB limit.

Task 5 – Image Sensing and Acquisition: Camera Arrangements [5 marks]

Prediction (before testing): Rolling-shutter area-scan with continuous light will blur and skew; global-shutter with a strobe freezes motion but is limited by field of view; line-scan gives the highest resolution across a 1.6 m web provided it is synchronised to an encoder.

Movement during exposure (v = 0.75 m/s = 0.75 mm/ms)

Exposure	Movement = v × t	Pixels at 0.13 mm/px	Compared with 0.5 mm defect	Peak contrast kept (simulated hole)
0.1 ms	0.075 mm	0.58	15% of defect size – acceptable	105 of 105 grey levels
0.5 ms	0.375 mm	2.88	75% of defect size – marginal	100
2 ms	1.500 mm	11.52	3× the defect – defect smeared to 1.5 mm	30 (only 28% retained)

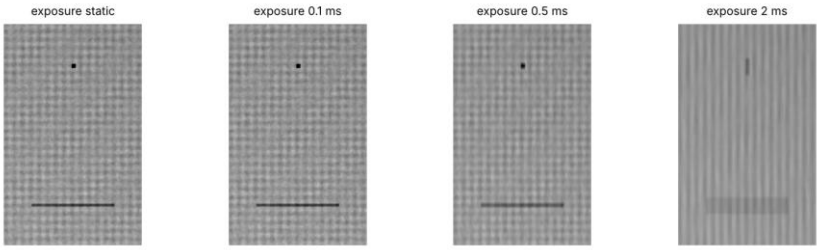


Figure 10 – Simulation (not a captured moving strip): motion blur along the travel direction for static, 0.1, 0.5 and 2 ms exposures.

Assumptions of the simulation: synthetic fabric, blur modelled as a box filter of length v×t along the travel axis, 0.13 mm/px; no lens blur, no sensor lag.

Line-scan line rate and encoder synchronisation

Sampling 0.2 mm along travel at 750 mm/s needs **line rate = 750 ÷ 0.2 = 3,750 lines/s** (period 0.267 ms; exposure ≤ 0.267 ms gives ≤0.2 mm blur, which is also the stage-6 choice of 3,840 lines/s for 0.195 mm). An encoder on a measuring roller (example: 100 mm roller, circumference 314.16 mm, 1,000 PPR × 4 quadrature = 4,000 counts/rev = 0.0785 mm/count) sends pulses; the camera's line-trigger divider takes one line every ≈2.55 counts (0.2 mm). Because lines are triggered by web distance, not time, speed changes do not distort the image, and the encoder count also gives the fault's position along the roll.

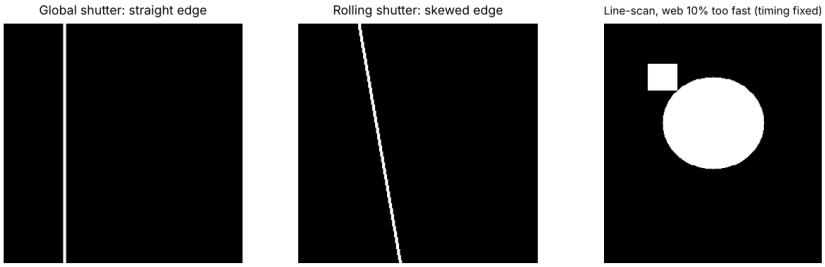


Figure 11 – Left: global shutter (straight). Middle: rolling-shutter skew (simulated, 30 µs per row). Right: line-scan image when web runs 10% fast with fixed line timing.

If speed changes by 10% but timing is not updated: each line then spans 0.22 mm (fast) or 0.18 mm (slow) instead of 0.2 mm, so the image is compressed by 9.1% (or stretched by 11%) along travel; a 0.5 mm hole is measured as ≈0.45 mm long, circles become ellipses, and fault positions drift. Encoder triggering removes this.

Comparison of the three arrangements

Aspect	A. Area rolling-shutter	B. Area global-shutter + strobe	C. Line-scan
Motion	Needs short exposure; continuous light → blur/skew (30 µs × 2,048 rows = 61 ms readout; 0.75 m/s → 46 mm if motion crosses rows)	All pixels exposed together; 50–100 µs strobe → ≤0.075 mm	Each line exposed 0.27 ms → ≤0.2 mm
Line-rate / frame demand	≈10 fps with overlap	Strobe-synchronised, frame rate set by overlap	3,840 lines/s (8k)
100 Hz flicker (fluorescent)	Row-wise brightness bands (period 7.5 mm along web at 0.75 m/s)	Strobe from DC supply: none	DC LED needed; with fluorescent: bands
Colour acquisition	Bayer: effective resolution ≈½–⅔; interpolation artefacts	Mono recommended; Bayer same issue	Mono (this design); tri-linear/3-sensor gives true colour at higher cost
Coverage of 1,640 mm	Several cameras (~4) to reach ≈0.2 mm/px	Same	One 8k camera, f ≈24 mm
Approx. cost (indicative)	Low	Medium–high (multi-camera + strobe controller)	Medium–high (one camera, light, encoder)

Sensor signal chain; CCD vs CMOS

Photons strike each photodiode and are converted (quantum efficiency ≈50–80%) into photo-electrons, collected during exposure. The charge is converted to a voltage by an amplifier (in CMOS, per pixel; in CCD, at the output after charge transfer), then **analogue conditioning** (gain, offset/black-level, correlated double sampling to remove reset noise), then an **ADC** (e.g. 10–12 bit) produces a digital number; after optional digital correction (gain, flat-field) it is written as the stored pixel value, commonly 8-bit. **CCD vs CMOS:** CCD gives very uniform output and low noise but is slower and costs more; modern CMOS is fast, cheaper, has lower power and per-column ADCs, and for line-scan at 3–6 kHz either works. For this budget **CMOS** is the practical choice; CCD line sensors would only be considered if very low noise were critical.

Recommendation and failure condition

Select C: line-scan, encoder-triggered, DC LED illumination. It meets 0.5 mm at 0.195–0.2 mm/px (Task 6), avoids rolling-shutter skew and mosaicking seams, and handles the full width with a single camera. **It fails** when the web speed or lateral position is not tracked: a roller slip, a stretched web, or a lateral shift beyond the ±8 mm field margin; also when exposure per line is too short for the lighting power (images become noise-limited, losing the low-contrast faults).

Task 6 – Sampling: Spatial Resolutions [5 marks]

Prediction (before testing): 4,096 px will show the 0.5 mm hole in ≈1 px and fail on size; 8,192 px will detect it but with size error ≈0.1 mm; 12,288 px will have the smallest size error and the best faint-fault detection.

Quantity (1,600 mm field of view)	A: 4,096 px	B: 8,192 px	C: 12,288 px
mm / pixel = 1600 ÷ N	0.391	0.195	0.130
Pixels across 0.5 mm defect	1.28	2.56	3.84
Line rate for square pixels (0.75 m/s)	1920 lines/s	3840 lines/s	5760 lines/s
Data rate (8-bit mono)	7.9 MB/s	31.5 MB/s	70.8 MB/s
% of Gigabit Ethernet (≈110 MB/s usable)	7%	29%	64%
7 days, 24 h, all lines raw	4.8 TB	19.0 TB	42.8 TB
vs 2 TB limit	2.4× over	9.5× over	21.4× over

Nyquist: to represent a feature of width w without aliasing the sampling interval must be ≤ w/2, i.e. at least **2 pixels** across 0.5 mm. That is a theoretical minimum for an ideal, perfectly aligned sampler; real sensors integrate over the pixel, features land at random phase, there is lens blur and noise, so **3–4 pixels** are needed to measure size and keep contrast. Option A (1.28 px) is under Nyquist, B (2.56 px) is just above, C (3.84 px) is in the practical range.

Resampling experiment

A synthetic reference fabric of 100 mm × 100 mm was rendered at 0.065 mm/px (supersampled), with a 0.5 mm hole and a 0.5 mm × 12 mm gap, then "acquired" at the three sampling pitches by area-averaging (a sensor integrates light over a pixel). Detection used a local-background residual with a robust (MAD) z-score. 30 random texture/noise realisations were run per cell.

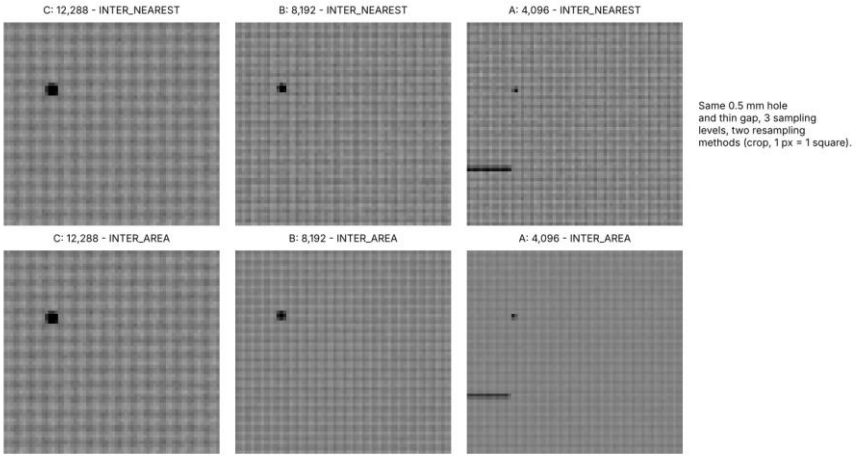


Figure 12 – Same hole and gap resampled from level C to B and A with INTER_NEAREST (top) and INTER_AREA (bottom).

INTER_NEAREST vs INTER_AREA: nearest-neighbour copies one source pixel per output pixel – it is fast but discards the others, producing jagged edges, aliasing (moiré on the 0.6 mm yarn pattern) and sometimes dropping a thin gap entirely. INTER_AREA averages all source pixels covered by the new pixel, like a real sensor, so thin features fade smoothly and texture does not alias. For simulating a sensor use INTER_AREA.

Test (30 trials per cell)	A: 4,096 px	B: 8,192 px	C: 12,288 px
0.5 mm hole (−85 grey)	100% (|size err| 0.21 mm)	100% (|size err| 0.06 mm)	100% (|size err| 0.07 mm)

Test (30 trials per cell)	A: 4,096 px	B: 8,192 px	C: 12,288 px
0.5 mm-wide thin gap (−52 grey)	100% (|size err| 0.22 mm)	100% (|size err| 0.10 mm)	100% (|size err| 0.02 mm)
Faint 0.5 mm hole (−25 grey)	27% (|size err| 0.06 mm)	37% (|size err| 0.27 mm)	50% (|size err| 0.27 mm)
Low-contrast shade band (−7 grey, 12 mm wide)	100% (|size err| 1.91 mm)	100% (|size err| 2.04 mm)	100% (|size err| 2.03 mm)
False alarms on 30 normal windows	0%	0%	0%

Size error for the shade band is its width error (12 mm true) and is small compared with its extent. Processing: the detector time scales with pixels; the optimised implementation (1,024 × 1,024 tile, 1 core) ran in 41 ms = **26 Mpx/s per core** (measured); 4 cores are *assumed* to give ≈3× (≈75 Mpx/s), so B (31.5 Mpx/s, ≈40% load) is workable and C (70.8 Mpx/s, ≈95% load) is at the limit.

Frame / line rate and overlap

Line rate = web speed ÷ sampling interval: 750 mm/s ÷ 0.195 mm = 3,840 lines/s (B). For an area-scan camera with a field length L along travel the frame rate must satisfy fps ≥ v ÷ (L − overlap). If fewer frames than this are taken, **gaps** appear in coverage and a fault passes unseen; overlap (typically 10–20%) guarantees that a fault straddling a frame edge appears complete in one frame, at the cost of 10–20% more data. In line-scan there is no overlap along travel; across the web a ±8 mm lateral shift is covered by a 24 mm margin on the 1,640 mm field of view.

Consequence of the cheapest option, and recommendation

Choosing 4,096 px (0.39 mm/px) to save cost leaves the 0.5 mm defect at 1.3 px: high-contrast holes are still found here (100%), but size error was 0.21 mm (≈40% of the defect), faint holes were found only 27% of the time, and any real blur or noise would pull the high-contrast result down too. **Recommendation: B – 8,192 px (0.195 mm/px).** It found 100% of strong 0.5 mm faults with 0.06 mm size error (same as C), keeps the data rate within Gigabit Ethernet and the CPU, and a 7-day policy within 2 TB (Task 4). C (12,288 px) found the faint 0.5 mm hole more often (50% vs 37%) but costs 2.2× the data and likely exceeds the budget; it is the first upgrade if the missed-defect target is not met.

Outcome vs prediction (differs): I predicted C would have the smallest size error; the measured size errors for B (0.06 mm) and C (0.07 mm) are the same within trial noise – size accuracy saturated once the hole covered ≥2.5 px, and the benefit of C appeared only for the faint hole.

Task 7 – Quantization: Intensity and Colour Bit Depths [4 marks]

Prediction (before testing): A 6–8 grey-level shade fault will vanish or be unreliable at 4–5 bits; 10/12-bit will not change detection in textured fabric but will help on smooth dyed areas.

Method: a floating-point "original" (kept unchanged) was re-quantized to 12, 8, 6, 5 and 4 bits (12-bit is simulated from the float original; no real 12-bit sensor data). A −6.5 grey-level (8-bit scale) shade band was added to (i) textured fabric and (ii) a smooth dyed region (σ=0.35 grey noise, 10-grey gradient). "Contrast" = mean difference between band and neighbouring cloth in 8-bit units; "SD" = spread of that estimate/background.

Bit depth (levels)	Textured: contrast / background SD	Smooth dyed: contrast / SD of band-minus-neighbour	Smooth region: distinct levels used
12-bit (4096)	6.82 / 8.2	6.50 / 0.07	294
8-bit (256)	6.82 / 8.2	6.51 / 0.09	20
6-bit (64)	6.81 / 8.3	6.25 / 1.54	6
5-bit (32)	6.80 / 8.5	6.80 / 2.65	3
4-bit (16)	7.01 / 9.6	6.82 / 7.97	2

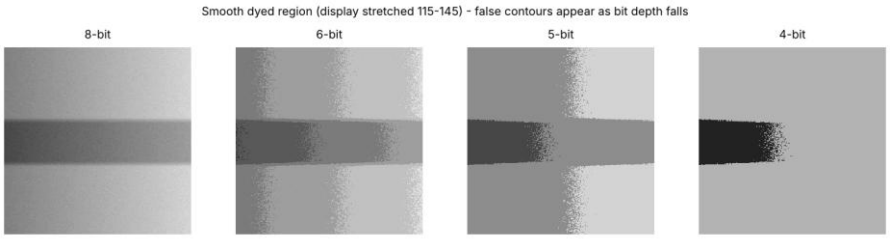


Figure 13 – Smooth dyed region re-quantized (display stretched 115–145): false contours appear at 6 bits and below.

Outcome vs prediction (differs in part). On **textured fabric** the mean contrast (6.8) survived even at 4 bits: the natural texture and noise (SD 8–10 grey) act as dither, so averages over many pixels still reveal the band. On the **smooth dyed region** the *average* contrast survived but its reliability collapsed: the variability of the band estimate rose from 0.07 (12-bit) to 0.09 (8-bit), 1.5 (6-bit) and 8.0 (4-bit), and at 5 and 4 bits only 3 and 2 levels remained. A gradual 10-grey gradient turns into visible steps: each step is a false contour that a detector can read as a shade boundary. So fewer levels do not erase the faint fault, they replace a smooth transition with steps and make the decision noisy. 8-bit is the lower limit; below it is unsafe for dyed smooth cloth.

Storage per image and for 7 days

Representation	8,192 × 4,096 frame	7 days of all lines (31.5 MB/s at 8-bit)
8-bit	33.6 MB	19.0 TB
12-bit (packed)	50.3 MB	28.5 TB
12-bit in 16-bit container	67.1 MB	38.1 TB
6-bit	25.2 MB	14.3 TB
5-bit	21.0 MB	11.9 TB
4-bit	16.8 MB	9.5 TB

Distinguish: **sensor bit depth** (ADC resolution, e.g. 12-bit), **file encoding** (8-bit PNG, 16-bit PNG, JPEG), and **model precision** (e.g. 32-bit float or 8-bit integer weights in a network). They are independent: a 12-bit sensor can be saved as 8-bit PNG, and a model may compute in float32 on 8-bit input. **An 8-bit display does not mean the sensor captured only 8 bits:** the sensor may digitise at 12 bits, the correction chain (black level, gain, flat-field) works at higher depth, and only the final output or display is 8-bit.

Sharing a fixed storage budget: the target faults differ. A 0.5 mm hole needs *pixels* (spatial resolution) and high contrast, so extra bit depth adds almost nothing; a 5–8 grey shade band is large (millimetres to centimetres) and needs *levels*, not pixels – it can be detected on 4–8× binned data. So spend resolution on the full-width 8k line and add depth **before** flat-field correction. **Recommendation:** capture at 10–12 bits, apply flat-field in that depth, store the evidence image as 8-bit; do not store 4–6-bit.

Task 8 – Image Formation Models: Optics and Lighting [5 marks]

> **Prediction (before testing):** Diffuse enclosed lighting gives the most uniform image; directional low-angle light raises raised-yarn contrast but also fold shadows and texture, and so is worse for uniformity and false alarms.

Projection and photometric models

Geometry (pinhole): $x = f \cdot X/Z$, $y = f \cdot Y/Z$. A 0.5 mm feature at Z = 700 mm with f = 23.6 mm has image size $0.5 \times f/(Z - f) = 17.5$ µm = 2.5 pixels of 7 µm. Longer focal length or shorter distance → larger image (finer sampling) but smaller field of view; a point closer to the camera (a fold) appears larger and shifted outward.

Photometry: $f(x,y) = i(x,y) \times r(x,y)$, where i is the illumination reaching the point and r the reflectance (0–1) of the fabric (a defect changes r, e.g. a hole or oil mark; a fold changes the angle and so effectively i). Brighter light raises i uniformly, not r; a glossy fibre has a strongly angle-dependent r (specular), so its highlight can saturate and hide a defect.

Uniformity across the 1,640 mm web (simulated profiles)

Arrangement	Peak-to-peak variation (% of mean)	Standard deviation (% of mean)
A. Ambient + standard lens	61% (≈ shed windows / lux changes)	21%
B. Diffuse LED + calibrated lens	51% ($\cos^4$ fall-off to the edges)	15%
C. Directional low-angle	96%	26%

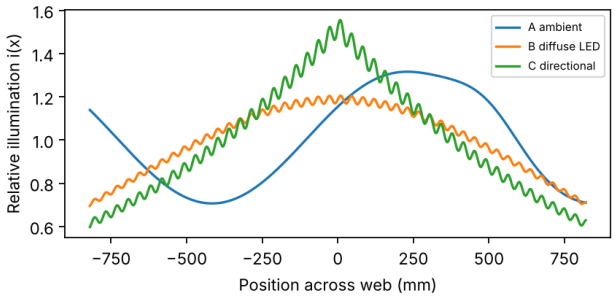


Figure 14 – Modelled illumination profile across the web for the three arrangements.

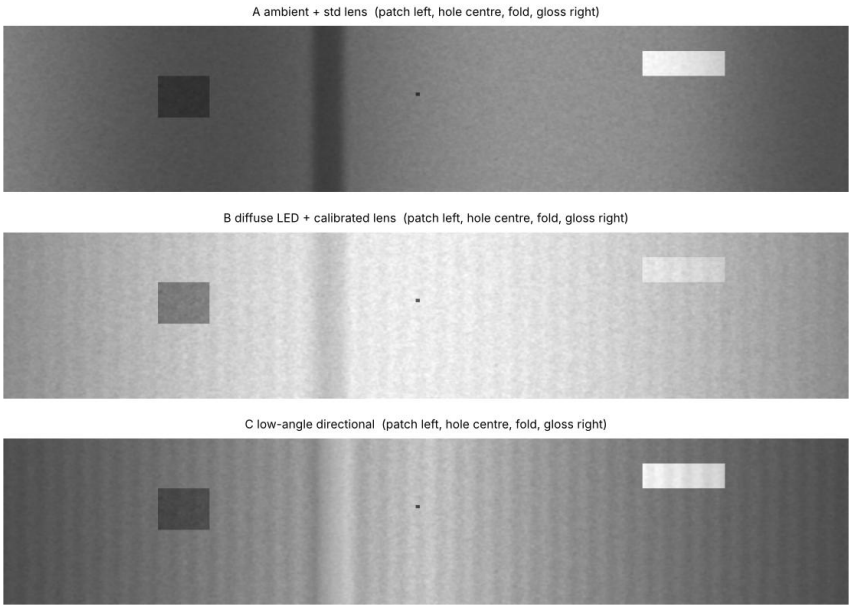


Figure 15 – Simulated strips: faint oil patch (left), small hole (centre), fold (~x=300), glossy patch (right).

Note: these profiles are **models with assumed parameters** ($\cos^4$ fall-off for the diffuse bar, exponential fall-off for the low-angle bar, window-shading pattern for ambient); they show relative behaviour. Absolute lux/uniformity needs a lux-meter or white-card capture on the real line. The glossy patch (right) saturates under ambient and directional light and is least disturbing under diffuse; the fold casts a shadow in all three, strongest for C.

Flat-field correction

Corrected = (raw − dark) ÷ (white reference − dark) × mean(white reference − dark). This assumes illumination varies **slowly** compared with fabric texture and defects (so the white-card image measures i alone), that the camera response is linear, and that r of the reference tile is uniform. It fails if lighting changes between the reference and the fabric (daylight), or on folds where i changes locally.

Arrangement	Across-web brightness variation before → after	Visibility of faint patch (contrast ÷ local noise) before → after
A ambient	23% → 6%	6.2 → 17.2
B diffuse LED	15% → 3%	4.1 → 16.9

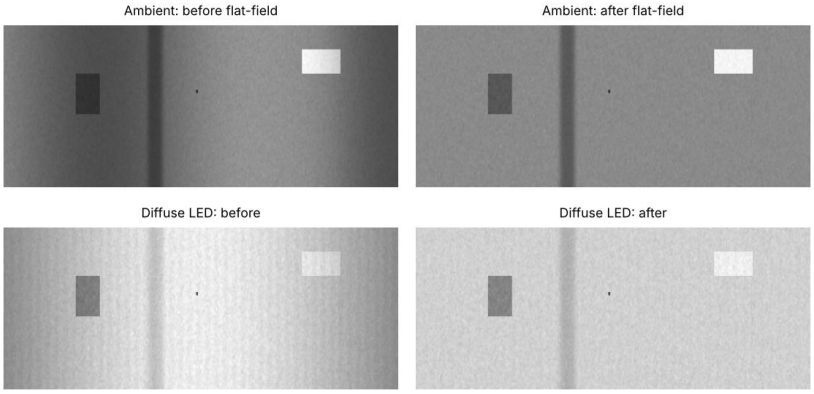


Figure 16 – Before/after flat-field correction (ambient and diffuse LED).

Flat-field reduces across-web brightness variation to ≈6% for the ambient case but, because it assumes a stable white reference, ambient daylight (1,500–45,000 lux) will keep it from being repeatable; the diffuse LED case drops to ≈3% and stays stable.

Focal length and field of view

Assumptions: line-scan sensor with 7 µm pixels (assumed), working distance WD = 700 mm, field of view = 1,600 mm + 2 × 8 mm lateral + 2 × 12 mm edge margin = **1,640 mm**. Magnification m = sensor length ÷ FOV; f = WD·m ÷ (1 + m).

Sensor	Sensor length	m	f (thin lens)	mm/px at the fabric
4,096 px	28.7 mm	0.0175	12.0 mm	0.400
8,192 px	57.3 mm	0.0350	23.6 mm	0.200
12,288 px	86.0 mm	0.0524	34.9 mm	0.133

Hence a **≈24 mm** lens for the 8k sensor. Its image circle must cover the 57 mm sensor, so a large-format line-scan lens (not a standard C-mount lens) is needed. Depth of field at F8 with a 2-pixel (14 µm) blur criterion ≈ $2 \cdot N \cdot c \cdot WD^2 / f^2$ ≈ 195 mm, so a few millimetres of web flutter are acceptable.

Perspective error from a raised fold

Fold height h	Scale error WD/(WD−h)	Error on a 20 mm feature	Position shift at web edge (820 mm off axis)
5 mm	0.72%	0.14 mm	5.9 mm
10 mm	1.45%	0.29 mm	11.9 mm

Fold height h	Scale error WD/(WD−h)	Error on a 20 mm feature	Position shift at web edge (820 mm off axis)
20 mm	2.94%	0.59 mm	24.1 mm

Position shift = x·h/(WD − h): a 10 mm fold at the web edge appears 12 mm out of place, large for a marker working to ±10 mm. A checkerboard-calibrated lens removes lens distortion and fixes mm/px **at the nominal plane only**; for off-plane points a **fixed web plane** (rollers, vacuum bed or tensioned span) keeps h near zero, and height cannot be recovered from a single camera. In line-scan the along-travel direction is not subject to this perspective error (one line, moving web), so only the across-web position shifts.

Recommendation

Select B: enclosed diffuse DC LED lighting with a calibrated ≈24 mm lens and a white-reference flat-field, on a fixed web plane. It gives the lowest across-web variation (51% peak-to-peak raw, 3% after correction), independence from daylight, and fewest false alarms (Task 9). Directional low-angle light (C) is kept as an optional second light for raised-yarn faults. **A condition that can still cause a false decision:** glossy synthetic fibres or a fold that reflects the diffuse light specularly into the camera; this produces a bright streak (Task 9: all 8 glossy and 8 fold test windows raised a flag at detector level and must be rejected by the classification stage).

Task 9 – Applications and Integrated System [6 marks]

9.1 Simulation set-up (measured vs estimated)

One common labelled dataset: 72 simulated 40 × 40 mm fabric windows rendered at 0.065 mm/px: 40 faulty (8 each of 0.5 mm hole, 0.5 mm gap, slub, oil mark, shade band) and 32 normal (8 each plain, twill, fold shadow, glossy streak). Each window is passed through an **acquisition model** for each configuration: lighting gradient and flicker, motion blur v×t, area-average sampling, sensor noise, flat-field residual, quantization and JPEG where used. The same detector (local background residual) is used for all, with the threshold *calibrated on 10 separate fault-free plain/twill images* to give no false alarms on texture. **Everything below in this section is a simulation; the configuration parameters are assumed.**

Parameter	A Entry	B Balanced	C High-detail
Sampling (1,600 mm)	4,096 px, 0.39 mm/px	8,192 px, 0.195 mm/px	12,288 px, 0.130 mm/px
Exposure → blur along travel	2 ms → 1.5 mm	0.267 ms → 0.2 mm	0.173 ms → 0.13 mm
Lighting	Ambient / fluorescent: ±15% gradient, 8% flicker	Diffuse DC LED, flat-field (1% residual)	Directional low-angle: texture ×1.5, gap ×1.3, slub ×2, oil ×0.6, shade ×0.7, fold ×2, gloss ×1.5 (assumed)
Noise / depth / format	σ=3.5 grey, 8-bit, JPEG q60	σ=2.5, 8-bit, lossless	σ=2.5, 10-bit, lossless

Prediction (before testing): C will find the most faults; A will miss small and low-contrast ones; B will be close to C but with fewer false alarms.

9.2 Results

Measure	A Entry	B Balanced	C High-detail
Faults detected (of 40)	5 (13%)	40 (100%)	37 (93%)
hole / gap / slub / oil / shade (of 8)	0 / 2 / 3 / 0 / 0	8 / 8 / 8 / 8 / 8	8 / 8 / 8 / 8 / 5
False alarms (of 32 normal)	6	16	16
plain / twill / fold / gloss (of 8)	0 / 0 / 0 / 6	0 / 0 / 8 / 8	0 / 0 / 8 / 8
Mean position error (compact faults)	1.52 mm	0.06 mm	0.04 mm
Detector threshold (calibrated; fine / coarse, in σ)	3 / 3	3 / 7	3 / 7
Processing time per 40×40 mm window (1 core, un-optimised detector)	3 ms	34 ms	107 ms
Stream rate (8-bit mono, estimated)	7.9 MB/s	31.5 MB/s	70.8 MB/s
7 days all lines raw (estimated)	4.8 TB	19.0 TB	42.8 TB
Storage with the Task 4 policy (estimated)	≈0.3 TB (JPEG)	≈0.4 TB	≈0.6 TB (10-bit, larger windows)
Latency budget (estimated, see 9.3)	≈0.4 s	≈0.46 s	≈0.6 s

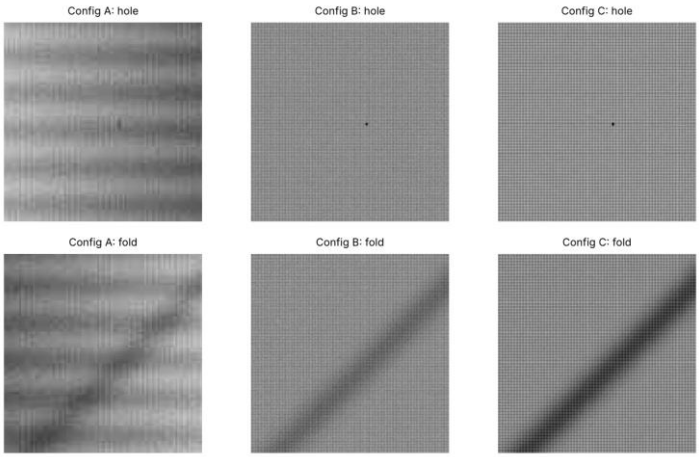


Figure 17 – Simulated acquisitions of a hole (top) and a fold (bottom) in configurations A, B, C.

Interpretation. Config A found only 5/40 faults: 2 ms exposure smears a 0.5 mm feature to 1.5 mm, and flicker and gradient bands (the flicker alone is ±10 grey) swamp the low-contrast faults; it also fired on 6/8 glossy streaks. B found **all 40** with 0.06 mm position error. C found 37/40: the directional light increases texture and fold contrast, which raised the calibrated detection threshold and made the low-contrast oil/shade faults harder (shade 5/8). At detector level **fold and gloss windows were flagged in B and C (16/32 each)** – the detector only finds anomalies; it is the Task 2 classification (shape, orientation, surrounding texture) that must reject those. That is a limitation: the false-alarm figure is for the detector alone, not for the full three-level pipeline. Robustness to folds is **not** achieved by hardware alone; lighting variation robustness is: B and C stay at 0 false alarms on plain and twill fabric.

Result that differs from my prediction: I expected C > B. In the simulation B ≥ C (40 vs 37 detections) because directional lighting reduces oil/shade contrast (×0.6–0.7) and enhances texture (×1.5); C's extra resolution (0.13 vs 0.195 mm/px) did not compensate. Limit of the test: the lighting multipliers are my assumptions, and a real directional light might be tuned to improve rather than reduce the oil and shade contrast; an actual test on fabric samples is needed before rejecting C's lighting.

9.3 End-to-end block diagram, time budget and design decisions

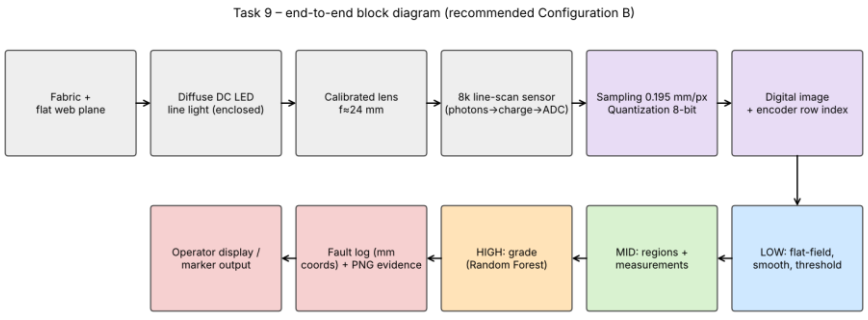


Figure 18 – End-to-end block diagram of the recommended configuration.

Stage (Configuration B, 0.75 m/s)	Time	How the estimate was made
Gather a 1,024-line block	0.267 s	1,024 lines × 0.195 mm = 200 mm ÷ 0.75 m/s (calculated)
Transfer over GigE	≈0.03 s	8.4 MB block ÷ ≈110 MB/s link (estimate; line-by-line transfer overlaps with capture)
Image processing (flat-field, residual, regions)	≈0.11 s	8.4 Mpx ÷ 75 Mpx/s (25 Mpx/s/core measured; ×3 on 4 cores assumed)
Classification + log write	≈0.01–0.02 s	RF predict + write of coordinates (measured 13.7 ms per image incl. features)
Marker signalling and communication	≈0.05 s	assumed PLC/IO delay

Stage (Configuration B, 0.75 m/s)	Time	How the estimate was made
Total	≈0.46 s	< 0.8 s required (≈0.34 s margin)

Decision	Evidence / Justification	What if not
Line-scan camera, 8,192 px mono	0.195 mm/px gives 2.56 px over 0.5 mm; 100% detection, 0.06 mm position error (Tasks 6, 9)	At 4,096 px: size error 0.21 mm and faint faults missed
Exposure ≤0.27 ms via encoder-timed lines	2 ms blur = 1.5 mm, 3× the defect (Task 5)	Blurred, undetected 0.5 mm defects
Enclosed DC LED diffuse light	Removes daylight/flicker; 3% residual non-uniformity after flat-field (Task 8)	Ambient: ≈6% residual, flicker bands, daylight-dependent
Flat-field correction with white reference	Raises faint-patch visibility 4.1 → 16.9 (Task 8)	Shade bands lost in the lighting gradient
Lossless PNG evidence, 8-bit	Bit-exact, ≈0.4 TB in 7 days (Task 4)	JPEG-only: measurements not reproducible; raw-all: 19 TB
Staged 3-level pipeline	0 false alarms on thick yarn vs 8/8 in rule-only (Task 2)	Rules only: many false alarms

9.4 Cost (indicative estimates – insert dated vendor quotations and citations)

No live vendor quotations were retrieved for this report. The prices below are order-of-magnitude **planning estimates in ₹** and are labelled as such; the brief requires current vendor quotations or catalogue prices with citations – please obtain these (e.g. from machine-vision distributors in India) and replace this table before submission.

Item (estimate, ₹)	A Entry	B Balanced	C High-detail
Camera	12 MP area-scan GigE: 35–45 k	8k mono line-scan GigE (budget-class): 90 k – 1.2 lakh	12k line-scan: 2.5–4 lakh
Lens	10–15 k	Large-format line-scan lens: 25–40 k	40–60 k
Lighting	Continuous LED 10 k	DC LED line light 1.7 m: 30–50 k	Directional + diffuse: 70 k+
Encoder, cabling, I/O	—	10–15 k	10–15 k
Enclosure / mounting	15–20 k	25–35 k	35–45 k
Total (estimate)	≈0.7–0.9 lakh	≈1.8–2.0 lakh (fits ₹2,00,000 if lower end holds)	≈4–6 lakh (does not fit)

B fits the budget only at the lower end of these estimates, so the quotation step is essential: if B's total exceeds ₹2,00,000, the first saving is a 4,096-px line-scan camera with a second lower-priced unit later, not removal of the encoder or enclosure.

9.5 How imaging fundamentals change in other applications

Aspect	Fabric inspection (this case)	Medical X-ray	Satellite observation	PCB inspection
Sensing band	Visible (400–700 nm), reflected	X-ray photons, transmitted	Visible to thermal IR, reflected/emitted	Visible, often plus UV/IR
Spatial resolution	0.2 mm/px over 1.6 m	0.1–0.2 mm/px detector pitch	0.3 m–30 m ground sampling	5–20 µm/px
Quantization	8-bit store, 10/12-bit sensor	12–16 bit (large dynamic range of tissue)	10–14 bit per band	8–10 bit
Image formation	Reflected light, controlled diffuse lamp	Absorption, geometry of point source	Sunlight + atmosphere, orbital geometry	Controlled multi-angle ring light

9.6 Final recommendation

Select Configuration B (balanced): 8k mono line-scan, encoder-timed, enclosed DC diffuse LED light, ≈24 mm calibrated lens, flat-field correction, 8-bit lossless PNG fault windows. **If the 2% missed-defect target is not met, upgrade first the spatial resolution (12k line-scan or a second camera)**, since the only weakness left in B (faint 0.5 mm hole: 37% detection vs 50% at 12k) is resolution-limited. Note the target

is not yet shown to be met: simulated detector-level recall is 100% on this set of 40, but with false alarms on folds/gloss still unresolved and with no real web trial.

Task 10 – Re-apply and Adapt the Imaging Front-End [8 marks]

10.1 Assumptions from Task 9 that no longer hold

Item	Task 9	New	Consequence if the old design is kept
Smallest defect	0.5 mm	0.2 mm hole + very faint contamination	0.2 mm at 0.195 mm/px is ≈1 px – below Nyquist
Web width	1,600 mm	1,800 mm	8,192 px → 0.22 mm/px (13% coarser)
Speed	0.75 m/s	1.2 m/s (+60%)	Line rate 5.5 kHz; blur 0.32 mm in 0.267 ms
Exposure	≤0.27 ms	≤0.09 ms for ≤0.11 mm blur	4× more light needed
Lighting	Enclosed, diffuse LED	Stronger daylight variation	Leakage/reflection through the enclosure, flat-field drift
Data volume	31.5 MB/s	45–179 MB/s depending on strategy	May exceed GigE and 4-core CPU
Reporting time	0.8 s	0.6 s	0.46 s budget at old speed grows because blocks shrink in time only if processing scales (see 10.3)

10.2 Recalculation

Sampling for a 0.2 mm feature: 2 px → 0.1 mm/px, 3 px → **0.067 mm/px**. For 1,800 mm that is 27,000 px across at 3 px; line rate = 1,200 ÷ 0.0667 = **18,000 lines/s**; data = 27,000 × 18,000 = **486 MB/s** – 4.4× Gigabit Ethernet and ≈6× the CPU capacity. Motion blur = v × t: at 1.2 m/s, 0.1 ms → 0.12 mm, 0.5 ms → 0.6 mm, 2 ms → 2.4 mm (compare 0.2 mm feature). Exposure must be ≤ ≈0.09 ms for blur ≤ 0.11 mm.

Strategy	Pixels across	mm/px	Px over 0.2 mm	Line rate	Data rate	% of GigE	7 days (24 h, all lines)
A Retain: 1 × 8k, FOV 1,800	8192	0.220	0.91	5461	45 MB/s	41%	27.1 TB
B Modify: 1 × 16k	16384	0.110	1.82	10923	179 MB/s	163%	108.2 TB
C Redesign: 2 × 8k, 2 NICs	16384	0.110	1.82	10923	179 MB/s	81% per NIC	108.2 TB

Data rate and CPU: the measured optimised detector gives ≈26 Mpx/s per core (1 core available in my test; ×3 on 4 cores is an assumption, ≈75 Mpx/s). Strategy A's 44.7 Mpx/s is ≈60% load – feasible. Strategy B/C's 179 Mpx/s is ≈2.4× the CPU capacity: the queue grows without limit and the decisions become **stale** (older than the 0.6 s deadline). Even storage cannot keep the full stream: 108 TB in 7 days. The Task 4 policy (windows around faults + thumbnail + periodic frames) keeps retention around 0.6 TB.

10.3 Compare strategies (calculations + small simulation)

Micro-hole simulation: a 0.2 mm hole at random sub-pixel position in synthetic fabric, 1.2 m/s motion blur included, 40 trials per row and 40 hole-free windows for false alarms; the back-light case models a bright hole against light transmitted through the fabric (assumed fabric transmission ≈12%, hole 90%, i.e. a large positive contrast).

Case	Detection of 0.2 mm hole	False alarms (hole-free windows)
A retain: top-light, 0.22 mm/px, 0.18 ms	50%	0%
B modify: top-light, 0.11 mm/px, 0.09 ms	93%	0%
C redesign: back-light, 0.11 mm/px, 0.09 ms	100%	0%
C2: back-light at old 0.22 mm/px	100%	0%

Faint contamination: a 4 mm diameter patch added to the same synthetic fabric with ±2.5 grey low-frequency dye mottling and detection at 0.11 mm/px: detection rate vs contrast = -2: 0%; -3: 0%; -4: 0%; -

6: 7%; -10: 20%; -25: 100% grey levels. Faint marks below about −10 grey are mostly invisible to this detector in normal colour images; they become reliably detectable at about −25 grey.

Criterion	A. Retain & optimise	B. Modify acquisition	C. Redesign sensing
0.2 mm micro-hole detectability	50% (top-light, 0.9 px) – fails	92% (top-light 0.11 mm/px)	100% with back-light, even at 0.22 mm/px
Colour / texture evidence	Unchanged	Better detail	Colour front light + extra spectral/UV test port
Faint contamination	Not detectable below ≈10 grey	Same	Detectable if a spectral band lifts contrast to ≈25 grey (must be tested on real samples)
Decision latency (0.6 s target)	Block 1,024 lines = 0.19 s + 0.12 s processing → ≈0.40 s ✓	Processing 179 Mpx/s ≫ 75 → stale ✗	Two streams → stale unless extra CPU ✗ (back-light variant with 1 camera: ≈0.40 s ✓)
Cost (indicative, ₹)	Lens + enclosure ≈0.5 lakh extra	16k camera + strobe ≈3 lakh+	2 cameras + backlight + NIC ≈4 lakh+ (single camera + back-light ≈1.2–1.4 lakh extra)
Storage 7 days (policy)	≈0.4–0.6 TB	≈0.6 TB	≈0.6–1 TB
Calibration / operational complexity	Low	Medium (strobe, alignment)	High (two light paths, registration)

10.4 Recommendation: MODIFY (staged), with a defined trigger for REDESIGN

The simulation shows that the dominant lever for **micro-holes** is not more pixels but **transmitted light**: a back-light turns a 0.2 mm hole into a bright point with contrast so high that it is found at 100% even with the existing 0.22 mm/px sampling (sub-pixel detection), whereas top-lit holes need 0.11 mm/px, which breaks the Gigabit link and CPU (179 MB/s). The full redesign (C) is therefore not needed for holes.
Recommended: MODIFY –

- **Sensor:** keep the 8k mono line-scan camera; run at 5.5 kHz (encoder-timed).
- **Lens:** change to ≈26 mm (FOV 1,840 mm; f = 700 × m/(1+m), m = 57.3/1,840) – the Task 9 24 mm lens is too wide-angle for the new width only by ≈10%; keep the large-format lens family.
- **Lighting:** add a line back-light under the web for hole detection; keep the DC LED front light, raise its output ≈4× for 0.09 ms line exposure, and enclose the imaging region with a light-tight hood against daylight (flat-field updated each shift).
- **Data format:** 8-bit mono, lossless PNG evidence windows; JPEG thumbnails for the continuous record; add the back-light channel as a second small window (not a second full stream).
- **Retained from Task 9:** PC, encoder, calibrated mounting, flat-field, three-level pipeline, storage policy.

Trigger for REDESIGN: if the lab test of the faint contamination marks (UV/NIR illumination on real samples) shows the contrast in normal colour stays below ≈10 grey, a spectral channel (UV/NIR light plus a suitable filter) must be added; this additional budget is not covered by ₹2,00,000 on my estimates and would need approval. Within the cap, the MODIFY option adds ≈₹1.2–1.4 lakh over the existing camera on indicative prices (to be replaced by quotations); whether this fits depends on whether the ₹2 lakh is counted for the new work only.

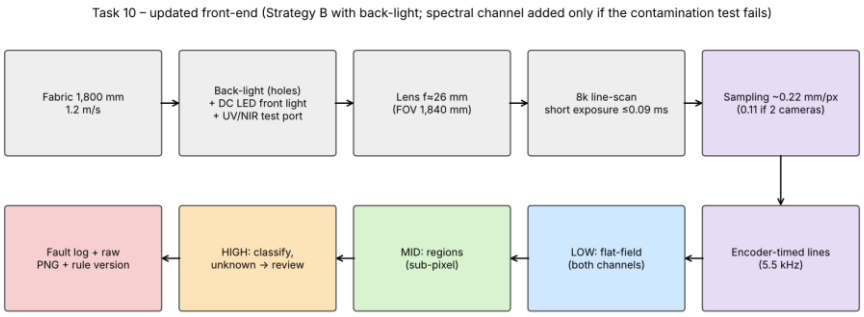


Figure 19 – Updated block diagram for the new requirements.

10.5 Time budget (1.2 m/s, 0.6 s target)

Stage	Time	Basis
Gather 1,024 lines (0.22 mm each = 225 mm)	0.19 s	calculated
Transfer + tile split	≈0.02 s	estimated
Processing 8.4 Mpx ÷ 75 Mpx/s	≈0.11 s	measured 25 Mpx/s/core, ×3 cores assumed
Classification, logging	≈0.02 s	measured
Marker/IO delay	≈0.05 s	assumed
Total	≈0.39 s	< 0.6 s (margin ≈0.2 s)

10.6 Operating condition where the redesigned system fails; monitoring test

Failure condition: a fabric with a **high-transmission open weave or a light colour** (or a backing carrier) reduces the back-light contrast; likewise any dust on the back-light window creates false bright specks and a thick/wet fabric blocks it. With transmission high enough that fabric ≈ hole in brightness, the 0.2 mm holes are lost. **Monitoring test before production:** run a **calibration fabric strip with laser-punched 0.2 mm holes and a printed set of faint marks of known contrast** through the line at 1.2 m/s at the start of every shift; the system must find ≥98% of holes and every contrast step above the target; a miss triggers the review procedure of Task 1. Also log the back-light white-reference level and alarm if it drifts >10%.

10.7 Adding a new fault category while preserving traceability

- Keep the **original** PNG evidence image, its encoder position and timestamp, never changing them.
- Store with each inspection a record: image hash, camera/light settings, **pipeline version and rule/model ID**, the feature vector, the decision, and the operator's disposition.
- Collect ≥30 labelled examples of the new fault from the review queue (Task 1), re-run the **stored** images through the new version to check old decisions do not silently change (regression set), and release as version N+1.
- Old records keep their old version number; a re-evaluation of old images with the new version is written as a separate, linked record, never an overwrite.